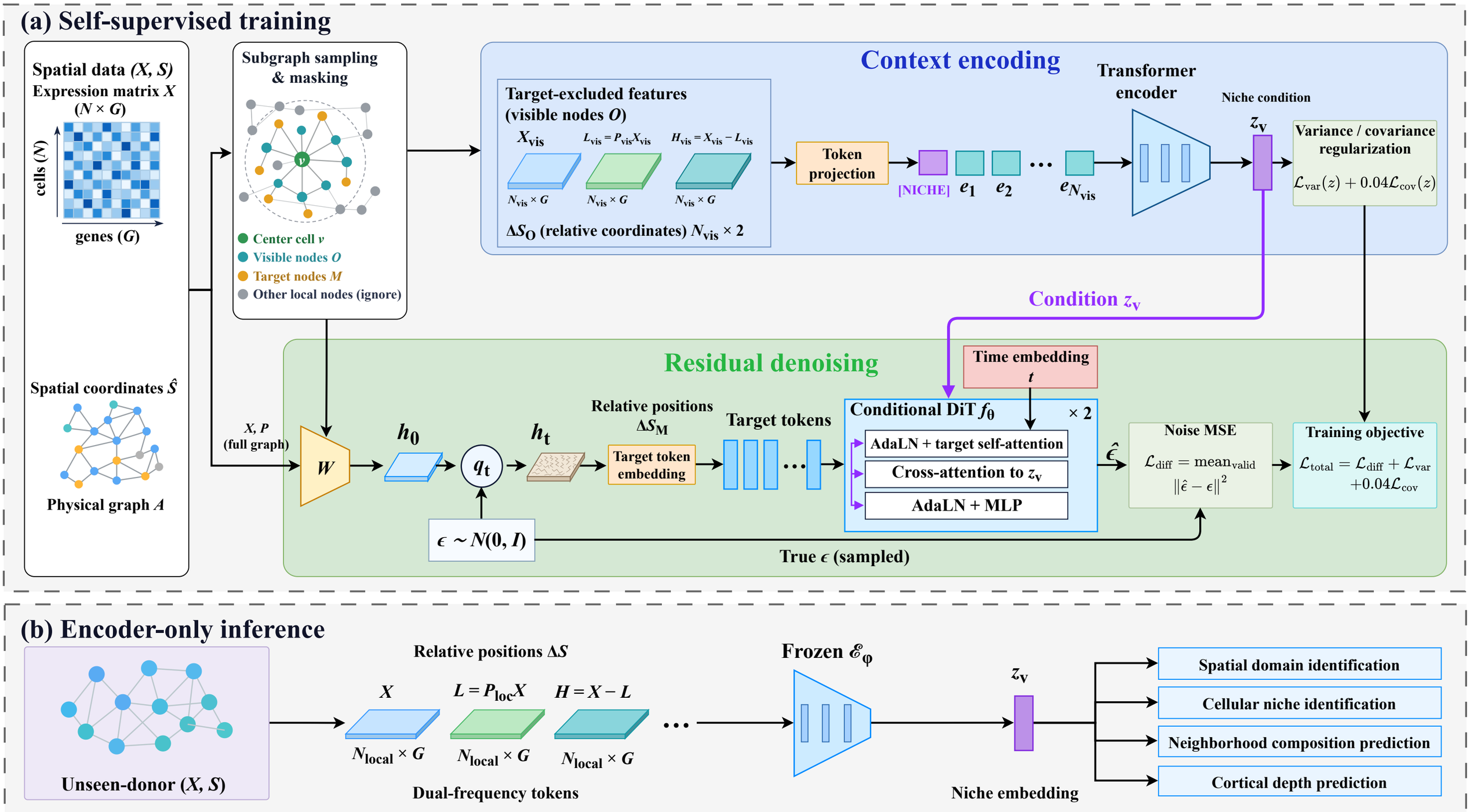

(a) Self-supervised training
Spatial data (X, S)
Expression matrix X
(N × G)
cells (N)
genes (G)
Spatial coordinates Ŝ
Physical graph A
Subgraph sampling
& masking
Center cell v
Visible nodes O
Target nodes M
Other local nodes (ignore)
Context encoding
Target-excluded features
(visible nodes O)
X_vis
L_vis = P_vis X_vis
H_vis = X_vis − L_vis
N_vis × G
ΔS_O (relative coordinates) N_vis × 2
Token projection
[NICHE] e_1 e_2 e_N_vis
Transformer
encoder
Niche condition
z_v
Variance / covariance
regularization
ℒ_var(z) + 0.04ℒ_cov(z)
Condition z_v
Residual denoising
Time embedding
t
X, P
(full graph)
W
h_0
q_t
h_t
Relative positions
ΔS_M
Target token
embedding
Target tokens
Conditional DiT f_θ
× 2
AdaLN + target self-attention
Cross-attention to z_v
AdaLN + MLP
ε̂
Noise MSE
ℒ_diff = mean_valid
‖ε̂ − ε‖²
Training objective
ℒ_total = ℒ_diff + ℒ_var
+0.04ℒ_cov
ε ~ N(0, I)
True ε (sampled)
(b) Encoder-only inference
Unseen-donor (X, S)
Relative positions ΔS
X
L = P_loc X
H = X − L
N_local × G
Dual-frequency tokens
Frozen ℰ_φ
z_v
Niche embedding
Spatial domain identification
Cellular niche identification
Neighborhood composition prediction
Cortical depth prediction